

Intelligent **Keyword** Extraction System

A Hybrid NLP Approach for Automatic Key Term & Phrase Identification

Abdul Rehman · Sohaib Maqsood · Usman Ali

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The Executive Summary

Developing a smart computer tool that automatically identifies core concepts from vast documentation. By blending semantic AI with statistical models, we turn hours of reading into seconds of insight.

| The Challenge: **Information** Overload

Data Explosion

Digital content is growing at an exponential rate.
Manually reading thousands of documents is impossible for modern organizations.

Method Weakness

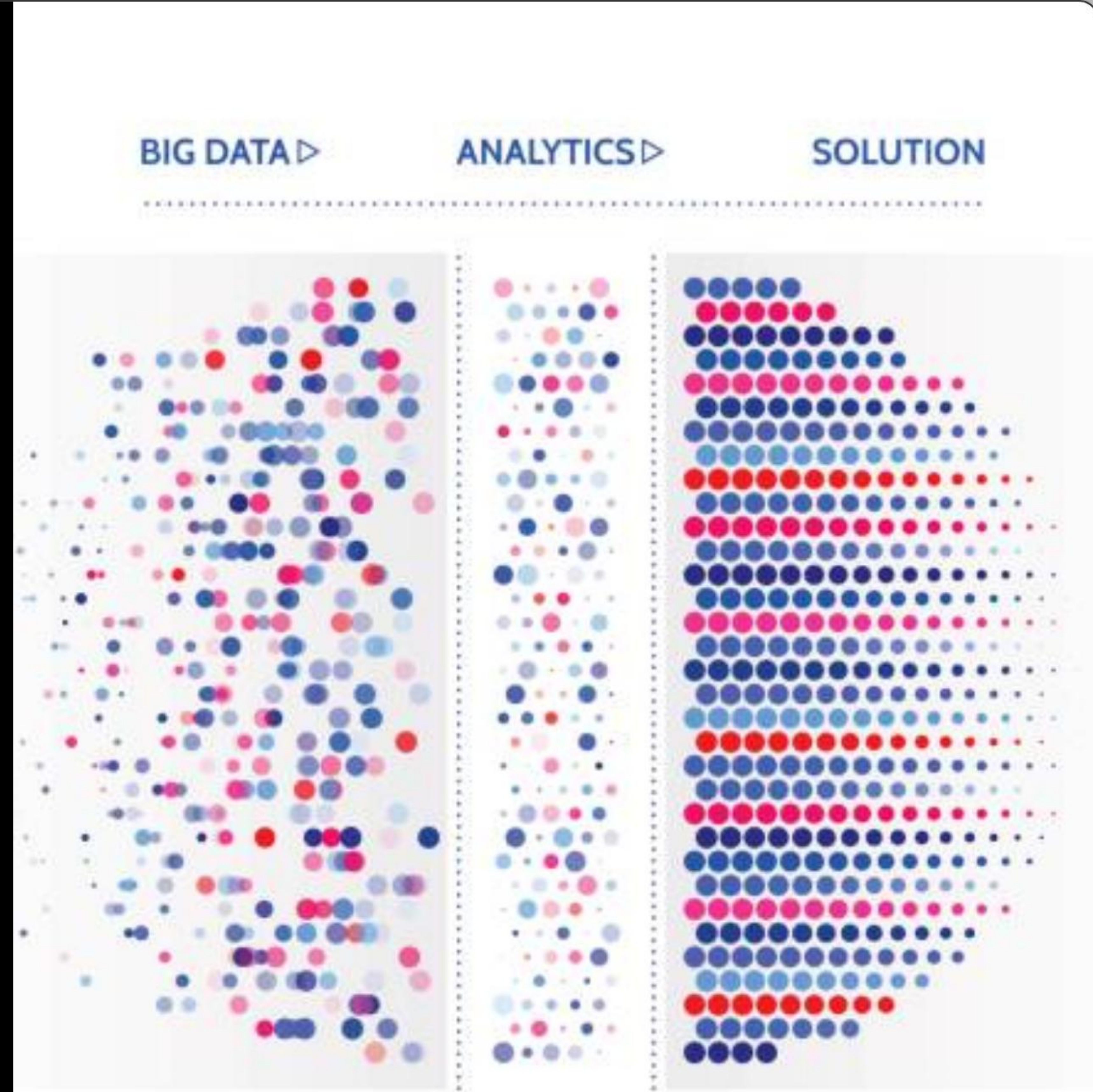
Standard tools rely on a single algorithm, often missing semantic context or failing at statistical frequency patterns.

"Existing solutions are slow, inconsistent, and lack powerful visualization."

| The **KWE** Solution

Our Intelligent Keyword Extraction System (KWE) is built on a "Hybrid Engine" architecture.

- ✓ High-speed document parsing (PDF, DOCX)
- ✓ Simultaneous multi-algorithmic extraction
- ✓ Dynamic Score Fusion & Visualization



| System Architecture Overview

01. Input Layer

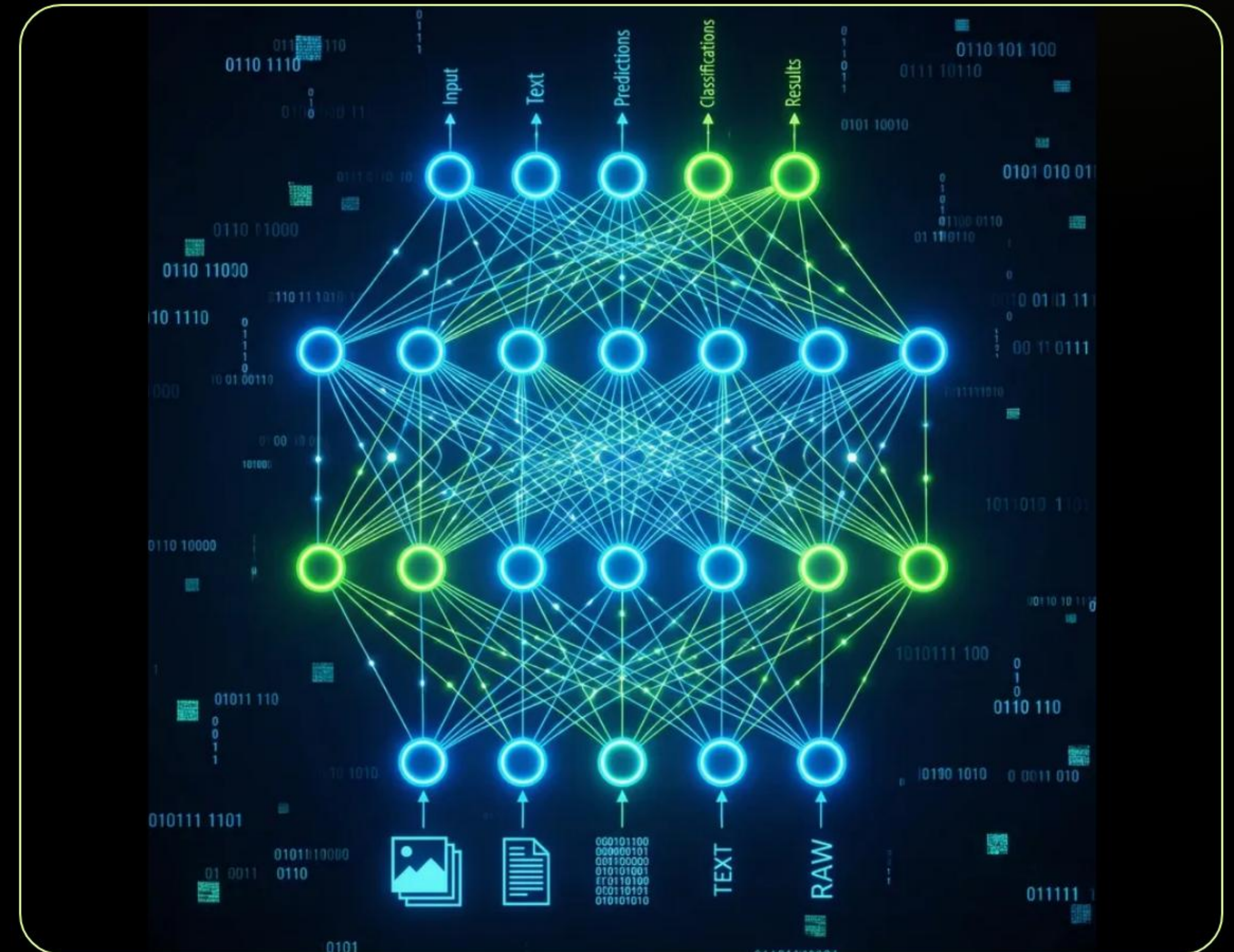
Clean text extraction from various file formats and noise removal preprocessing.

02. Processing Layer

Parallel execution of KeyBERT, RAKE, YAKE, and TF-IDF modules.

03. Output Layer

Fusion algorithm ranks terms for final presentation via word clouds and bar charts.



| The Hybrid Engine



KeyBERT

Semantic similarity
using BERT
embeddings.



RAKE

Statistical co-
occurrence of phrases.



YAKE

Linguistic feature
analysis
(position/casing).



TF-IDF

Classic importance
weighting (TF & IDF).

| Semantic vs. Statistical

KeyBERT (Semantic)

Uses Transformer models to understand the 'meaning' of words. It maps phrases into a high-dimensional vector space and finds terms that best represent the entire document's context.

RAKE (Statistical)

Rapid Automatic Keyword Extraction focus on word co-occurrence. It identifies phrases that appear frequently together while ignoring stop-words, optimized for speed.

| Linguistic & Term Importance

YAKE Feature Scoring

Scores words based on:

- Casing (Capitalized = Proper Noun?)
- Word Position (Intro/Conclusion weight)
- Term Relatedness to Context

TF-IDF Logic

Weights a term by its document frequency vs. corpus rarity.

$$W_{i,j} = tf_{i,j} \times \log \left(\frac{N}{df_i} \right)$$

| The Pipeline Workflow



| The Hybrid Advantage



**Accuracy based on Human-Annotated Keyword Benchmarks*

| Strategic Applications

Domain	Use Case	Impact
Digital Marketing	SEO & Competitor Research	Higher Search Visibility
Legal Review	Contract Clause Extraction	90% Time Savings
Academic Research	Literature Gap Analysis	Thematic Clarity
Business Intel	Customer Feedback Mining	Actionable Insight

| Market Potential & Roadmap

SaaS

Platform Subscription

Future Strategy

- **Multi-language:** Support for Urdu, Arabic, & Spanish.
- **API Services:** Microservice for external developers.
- **Cloud Integration:** One-click G-Drive/Dropbox sync.

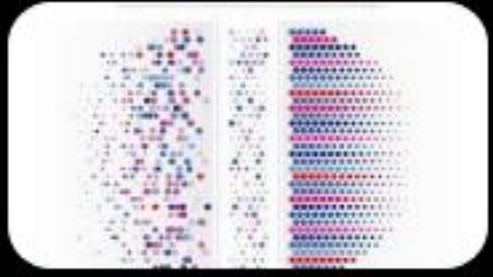
Questions ?

Turning Information Into Intelligence

`KWE.System@university.edu.pk`

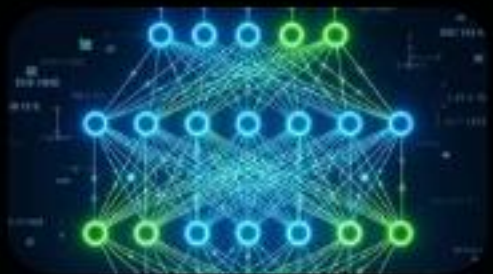
Project Repository: `github.com/KWE-Hybrid-NLP`

| Image Sources



<https://media.istockphoto.com/id/837344630/vector/big-data-visualization-abstract-geometric-background-with-letter-code-and-hexagons-technology.jpg?s=612x612&w=0&k=20&c=WasUZMz4UbxzIWIAEyudwtWznoT6HS0yOm-sY2qyRsQ=>

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